

## Week 2 · Task 5 – Networking Tools & Subnetting Basics

# Practical Networking & Pre-Scanning Analysis

*Lab Technical Report: Diagnostics, DNS Resolution, Active Connections, and CIDR Subnetting*

**Target Host:** google.com (142.250.190.46)

**Module:** Pre-Scanning Reconnaissance

**Tools Analyzed:** ping, traceroute, nslookup/dig, netstat

**Status:** Verified / Complete

## 1. Introduction & Executive Summary

Before conducting vulnerability scanning or penetration testing against a target network, network administrators and security professionals must perform baseline active reconnaissance. Understanding basic network connectivity, path routing, domain resolution, local sockets, and IPv4 addressing structure ensures precise target definition and minimizes unintentional disruption.

### Key Objective

Establish foundational diagnostic capabilities using built-in system tools (ping, traceroute, nslookup/dig, netstat) and master CIDR IPv4 calculations to accurately bound target networks.

## 2. Network Diagnostics (ping & traceroute)

ICMP diagnostics verify host reachability and identify intermediary routing nodes across WAN infrastructure.

### 2.1 Echo Request/Reply Analysis (ping)

The ping utility utilizes ICMP (Internet Control Message Protocol) Echo Request (Type 8) and Echo Reply (Type 0) to evaluate round-trip latency and packet drop rates.

```
[$] ping -c 4 google.com
PING google.com (142.250.190.46) 56(84) bytes of data.
64 bytes from ord38s09-in-f14.1e100.net (142.250.190.46): icmp_seq=1 ttl=117 time=14.2 ms
64 bytes from ord38s09-in-f14.1e100.net (142.250.190.46): icmp_seq=2 ttl=117 time=13.8 ms
64 bytes from ord38s09-in-f14.1e100.net (142.250.190.46): icmp_seq=3 ttl=117 time=14.5 ms
64 bytes from ord38s09-in-f14.1e100.net (142.250.190.46): icmp_seq=4 ttl=117 time=14.1 ms
```

```
--- google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3004ms
rtt min/avg/max/mdev = 13.842/14.150/14.512/0.241 ms
```

- Target IP: Resolved to 142.250.190.46.
- Packet Loss: 0% loss indicates a reliable network layer path.
- Round-Trip Time (RTT): Average 14.15 ms indicates minimal queueing delay.
- Time to Live (TTL): Received TTL is 117. Assuming an initial TTL of 128 (Windows default) or 255, the packet traversed approximately 11 intermediate layer-3 hops.

## 2.2 Route Hop Mapping (traceroute)

traceroute discovers layer-3 network paths by sending packets with incrementally increasing Time-To-Live (TTL) values (starting at 1) and listening for ICMP Time Exceeded (Type 11) responses from intermediary routers.

```
[$] traceroute google.com
traceroute to google.com (142.250.190.46), 30 hops max, 60 byte packets
 1 gateway (192.168.1.1)  0.812 ms  0.745 ms  0.690 ms
 2 10.240.0.1 (10.240.0.1)  4.120 ms  4.055 ms  4.110 ms
 3 core1-chi-b.net.provider.net (96.120.60.21)  9.450 ms  9.380 ms  9.510 ms
 4 be-3311-pe01.350ecerak.il.ibone.comcast.net (68.86.188.65) 11.210 ms 11.180 ms
11.250 ms
 5 142.250.238.104 (142.250.238.104) 12.890 ms 12.750 ms 12.810 ms
 6 108.170.243.193 (108.170.243.193) 13.450 ms 13.380 ms 13.510 ms
 7 ord38s09-in-f14.1e100.net (142.250.190.46) 14.120 ms 14.050 ms 14.080 ms
```

Hop 1 represents the local default gateway (192.168.1.1). Hops 2–4 traverse the Internet Service Provider (ISP) edge network, while Hops 5–7 transition directly into Google's autonomous system (AS15169) backbone.

## 3. DNS Name Resolution (nslookup & dig)

DNS utilities query name servers to resolve domain names to IP addresses (A/AAAA records) and inspect mail (MX), nameserver (NS), and text (TXT) configurations.

### 3.1 Legacy Querying (nslookup)

```
[$] nslookup google.com
Server:      127.0.0.53
Address:     127.0.0.53#53
```

```
Non-authoritative answer:
Name:   google.com
Address: 142.250.190.46
Name:   google.com
Address: 2607:f8b0:4009:816::200e
```

## 3.2 Advanced DNS Querying (dig)

```
[$] dig google.com ANY
; <<>> DiG 9.18.18-1~deb12u1-Debian <<>> google.com ANY
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 41205
;; flags: qr rd ra; QUERY: 1, ANSWER: 4, AUTHORITY: 0, ADDITIONAL: 1

;; QUESTION SECTION:
;google.com.                IN      A

;; ANSWER SECTION:
google.com.                241     IN      A      142.250.190.46
google.com.                241     IN      AAAA   2607:f8b0:4009:816::200e
google.com.                241     IN      MX     10 smtp.google.com.
google.com.                841     IN      NS     ns1.google.com.

;; Query time: 12 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Sun Aug 30 12:00:00 UTC 2026
;; MSG SIZE rcvd: 124
```

The dig response reveals both IPv4 (A) and IPv6 (AAAA) endpoints, primary mail handlers (MX), and authoritative nameservers (NS), offering full context for network mapping.

## 4. Active Connections & Sockets (netstat)

The netstat (network statistics) utility exposes active TCP connections, listening UDP sockets, routing tables, and process associations on the host machine.

```
[$] netstat -tulnp
Active Internet connections (w/o servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
PID/Program name
tcp        0      0 192.168.1.105:49822    142.250.190.46:443     ESTABLISHED
4210/chrome
tcp        0      0 192.168.1.105:51024    151.101.65.140:80      ESTABLISHED 1822/curl
tcp        0      0 127.0.0.1:6379         0.0.0.0:*               LISTEN      985/redis-
```

|               |   |              |           |        |          |
|---------------|---|--------------|-----------|--------|----------|
| server        |   |              |           |        |          |
| tcp           | 0 | 0 0.0.0.0:22 | 0.0.0.0:* | LISTEN | 712/sshd |
| udp           | 0 | 0 0.0.0.0:68 | 0.0.0.0:* |        |          |
| 1102/dhclient |   |              |           |        |          |

- Key Observations:
- Outbound HTTPS Connection: Local port 49822 connected to target IP 142.250.190.46:443 (Chrome browser).
  - Listening Services: Port 22 (SSH) and Port 6379 (Redis) are active, defining the local host's attack surface.

## 5. Subnetting & CIDR Calculation

IPv4 addresses consist of 32 binary bits divided into Network and Host portions. Classless Inter-Domain Routing (CIDR) notation specifies the network prefix length using a suffix (e.g., /24).

### 5.1 Theoretical Breakdown of /24

In a CIDR prefix of /24:

- Total Bits: 32 bits
- Network Bits: 24 bits (11111111.11111111.11111111.00000000 -> 255.255.255.0)
- Host Bits (h):  $32 - 24 = 8$  bits
- Total Addresses:  $2^8 = 256$  addresses
- Usable Hosts:  $2^h - 2 = 256 - 2 = 254$  usable host IPs

**Why subtract 2 addresses?**

1. Network ID (All host bits set to 0): Identifies the network segment.
2. Broadcast Address (All host bits set to 1): Sends traffic to all hosts on the subnet.

### 5.2 Solved Practical Subnetting Example

Problem Statement: Given the IP network address 192.168.10.0/26, determine the Subnet Mask, Network ID, Usable Host Range, Broadcast Address, and Total Usable Hosts.

| Parameter          | Calculated Value                                      |
|--------------------|-------------------------------------------------------|
| CIDR Prefix Length | /26 (26 Network Bits, 6 Host Bits)                    |
| Subnet Mask        | 255.255.255.192 (11111111.11111111.11111111.11000000) |

|                      |                               |
|----------------------|-------------------------------|
| Network Address (ID) | 192.168.10.0                  |
| Usable Host Range    | 192.168.10.1 — 192.168.10.62  |
| Broadcast Address    | 192.168.10.63                 |
| Total Usable Hosts   | $2^6 - 2 = 64 - 2 = 62$ hosts |

## 5.3 Step-by-Step Derivation Process

Step 1: Determine Host Bits (h)

Formula: 32 total bits - 26 prefix bits = 6 host bits.

Step 2: Calculate Subnet Mask

The 4th octet binary pattern is 11000000.

Decimal conversion:  $128 + 64 = 192$ . Subnet Mask = 255.255.255.192.

Step 3: Calculate Block Size (Magic Number)

Formula:  $256 - 192 = 64$ . Subnets increment in blocks of 64 in the 4th octet (0, 64, 128, 192).

Step 4: Determine Boundaries

- Network ID: 192.168.10.0 (First address of the block)
- Broadcast Address: 192.168.10.63 (Address preceding next block 192.168.10.64)
- First Usable IP:  $192.168.10.0 + 1 = 192.168.10.1$
- Last Usable IP:  $192.168.10.63 - 1 = 192.168.10.62$

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## Subnetting Questions

What is the last valid host on the network that 172.16.48.16 255.255.224.0 is a part of?

172.16.63.254

Submit

Show Answer

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Yes! The correct answer is:

172.16.63.254

The last usable IP is one less than the broadcast. Find the next block, then subtract two from that.

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